

Homework 6

Instructions:

- Take the acceleration due to gravity to be $g = 9.80 \text{ m/s}^2$
- Express all numerical results to two decimal places

Please submit your answer to the Web Learning by May 10th.

1. Relativistic Signal Reflection

A spaceship equipped with a radio transmitter and receiver is moving away from Earth at a speed of $u = \frac{4}{5}c$. After an astronaut sends out a radio signal, the signal is reflected by Earth and is received by the astronaut 60 s later.

- (a) At the instant the signal is reflected by Earth, how far is Earth from the spaceship as measured in the spaceship's frame of reference?
- (b) When the reflected signal reaches the spaceship, how far is the spaceship from Earth as measured in the Earth's frame of reference?

2. Relativistic Velocity Addition

One cosmic-ray particle approaches Earth along Earth's north-south axis with a speed of $0.80c$ toward the geographic north pole, and another approaches with a speed of $0.60c$ toward the geographic south pole. What is the relative speed of approach of one particle with respect to the other?

3. Nuclear Binding Energy of Deuteron

The rest mass of a proton is $m_p = 1.67265 \times 10^{-27} \text{ kg}$, and the rest mass of a neutron is $m_n = 1.67495 \times 10^{-27} \text{ kg}$. The rest mass of a deuteron (deuterium nucleus) formed by the combination of a proton and a neutron is $m_D = 3.34365 \times 10^{-27} \text{ kg}$.

- (a) How much energy is released during this combination process, in MeV?
- (b) This energy is known as the binding energy of the deuteron. What percentage of the deuteron's rest energy does this binding energy represent?

4. Compton Scattering of a Photon

A photon with frequency ν has energy and momentum given by $h\nu$ and $h\nu/c$, respectively, where h is the Planck constant. In a laboratory experiment, a photon with frequency ν_0 undergoes an elastic collision with a stationary free electron. After the collision, the photon's direction of travel is deflected by an angle θ relative to its original incident direction. Given that the rest mass of the electron is m_0 , find the frequency of the photon after the collision, ν .

5. Kinematics of Simple Harmonic Motion

A system consisting of a small ball and a light spring oscillates according to the law:

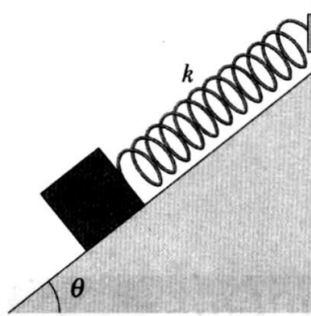
$$x = 0.05 \cos\left(8\pi t + \frac{\pi}{3}\right) \quad (\text{SI units})$$

- (a) Find the angular frequency ω , period T , amplitude A , initial phase ϕ_0 , maximum velocity v_{max} , and maximum acceleration a_{max} of the oscillation.
- (b) Find the phase of the oscillation at times $t = 1 \text{ s}$, 2 s , and 10 s .
- (c) Sketch the curves of displacement, velocity, and acceleration versus time.

6. Spring-Mass System on an Incline

As shown in the figure, a block of mass m can slide without friction on an incline at an angle θ . It is connected to the top of the incline by a massless spring with an unstretched length L_0 and a spring constant k .

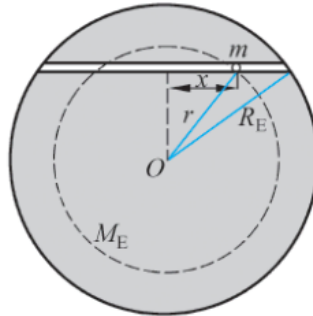
- How far from the top of the incline is the block's equilibrium point?
- If the block is pulled slightly down the incline and released, what is the period T of the resulting oscillations?



7. Oscillation in a Gravity Tunnel

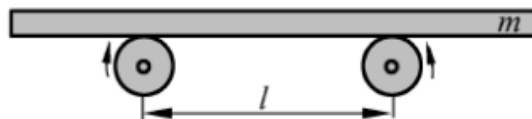
Imagine a straight, narrow, frictionless tunnel dug through the Earth. A ball of mass m is placed inside the tunnel at a distance x from the center of the tunnel. Assume the Earth is a uniform sphere with mass M_E and radius R_E .

- Find the gravitational force acting on the ball. (Hint: The ball is only affected by the gravitational pull of the mass within the sphere interior to the ball's position.)
- Prove that the motion of the ball in the tunnel under gravity is simple harmonic motion, and find its period T .



8. Plank on Counter-Rotating Rollers

As shown in the figure, a uniform long plank of mass m is placed symmetrically on two rollers separated by a distance $l = 20$ cm. As shown in the figure, the two rollers rotate in opposite directions. It is known that the coefficient of friction between the roller surfaces and the plank is $\mu = 0.5$. If the plank is moved a certain distance along the horizontal direction and then released, prove that the plank will undergo simple harmonic motion thereafter and find its period.



9. Rolling Cylinder with Spring Attachment

As shown in the figure, a solid cylinder of mass M attached to a horizontal spring (spring constant k) rolls without slipping along a horizontal surface. If the system is released from rest when the spring is stretched by a distance A , find:

- (a) the translational kinetic energy and
- (b) the rotational kinetic energy of the cylinder as it passes through the equilibrium position.
- (c) Show that under these conditions the cylinder's center of mass executes simple harmonic motion and find the period T .

(**Hint:** Find the time derivative of the total mechanical energy.)

